

FRAUD, BONDS AND INSURANCE

Making lies expensive

How fraud is stopped on attested data, whether a bond market can work when someone has to put up capital, and what a native insurer would realistically be.

THE STARTING POINT

Attestation proves provenance, not truth

INFORMATIONAL**Corroboration**

A fact counts when independent sources with different incentives agree, and cash is the final arbiter. Cheap to scale on structured data; the platform's native strength.

ECONOMIC**Bonds and holdbacks**

Capital at risk behind a claim, released or forfeited on a later mechanical truth. Works for facts with a verifiable outcome; useless for opinions.

PROFESSIONAL**Licence and liability**

A notary's office, an auditor's opinion, a valuer's signature: stakes that cannot be sold and take years to earn. The strongest deterrent and the one that stays off-chain.

Insurance prices what survives all three. Every residual risk is carried by someone; today that is W&I, title and professional indemnity cover. The better the three layers, the cheaper the residual.

Consensus is not on this list. Chain consensus orders transactions nodes can all see. It cannot decide whether a tenant exists. The applicable primitive is a quorum of attesters, not proof of stake.

WHERE FRAUD HAPPENS

Six vectors, in order of how often they occur

Vector	How it is done	Caught today by	What breaks it on the platform
Fabricated source data	Phantom tenants, side letters, inflated rents	Buyer DD, months later	Cash reconciliation, multi-source quorum
Settlement diversion	Changed payment instructions by email	Often not at all	Instructions from the closing set only
Identity and authority	Fake counterparty, agent beyond permission	KYC at onboarding	Know Your Agent, scoped permissions, action log
Market manipulation	Fake mandates, spoofed boxes, collusive bids	Reputation, slowly	Committed equity, capacity proof, memory
Collusive valuation	A mark that suits seller and lender	Rarely	Independent signature, outlier detection
Internal and adversarial	Tuned models, injected documents	Audit, after the fact	Maker-checker, source-anchored extraction

The first two are the money. Misstated income and diverted closing funds account for most real estate fraud loss. Both have a mechanical truth the platform can see: cash.

The last two are the new ones. Agents reading documents can be steered by documents. Ingested content is data, never instruction; every extracted field cites the span it came from.

Ordering is qualitative, from transaction practice, not measured.

THE LAYERS

Six layers, each stopping what the one below cannot

Layer	Mechanism	Stops	Does not stop
0 Identity	KYC, Know Your Agent, scoped permissions	Impersonation, agents acting beyond mandate	A real party lying
1 Provenance	Proof of Ingestion, lineage, timestamps	Silent edits, undated data, deniability	False data entered at source
2 Corroboration	Multi-source quorum, cash reconciliation	Single-source fabrication	Conspiracy across all sources
3 Economic	Bonds, holdbacks, sureties with objective triggers	Cheap lies about verifiable facts	Lies about opinions; lies bigger than the bond
4 Professional	Independent acts: notary, valuer, auditor, depositary	Platform self-dealing, collusive marks	A corrupt professional
5 Memory	Counterparty record, permission log, immutable trail	Repeat offenders, insider reads	The first offence

Read the last column as the design brief. Each gap is closed by the next layer; the residual after layer 5 is what insurance prices. No single layer is asked to do everything.

Layers 0, 1, 2 and 5 are already in the stack. They are the validation engine and Know Your Agent. Layers 3 and 4 are the new decisions this deck is about.

ASSURANCE LEVELS

Confidence becomes a level a counterparty can require

Level	What sits behind the value	Who accepts it
A0	Self-declared, no lineage	Nobody on the platform
A1	Attested from one source system	Screening, early ranking
A2	Two independent sources agree within tolerance	Mandate scoring at full weight
A3	Reconciled to cash on the payment rails	Credit boxes fire indicative terms
A4	A3 plus a bond or surety with an objective trigger	Firm term sheets, held pairs
A5	A4 plus an independent signed act or an insurance policy	Settlement, LP reporting

REPLACES FRESHNESS ALONE

The confidence factor $c(t)$ in the mandate mechanics becomes a function of level and age. A fresh A1 ranks below an older A3.

PER FIELD, NOT PER ASSET

NOI can be A3 while capex is A1. A credit box asks for the level it needs on the fields it prices.

Level names are placeholders; the ladder is the point.

LAYER 2 · CORROBORATION MECHANICS

A quorum of sources, with cash as the tie-break

01 Independence is about incentive, not system. Two exports from the seller's ERP are one source. The tenant, the bank and the tax filing are three.

02 Agreement has a tolerance. Sources agree when they fall inside a band set per field: tight for rent, wider for capex estimates.

03 Cash outranks documents. Where a ledger and receipts disagree, receipts win and the ledger's source loses standing.

04 Disagreement is a signal. A field that loses quorum drops a level and flags the asset; it does not silently average.

WORKED: NOI ON ONE ASSET

illustrative

Source	Annual rent	Incentive
Seller's rent roll	€1,020k	Overstate
Property manager ledger	€1,020k	Follows seller
Bank receipts, 12 months	€958k	None
Tenant confirmations	€955k	Understate
Quorum value, tolerance 2%	€956k	A3

The rent roll and ledger agree with each other and with nothing else. Two dependent sources do not make a quorum; the asset is flagged and the seller's data source drops in standing.

Numbers are a worked illustration.

A bond is a holdback with rules instead of lawyers

Element	Rule	Illustrative
Who posts	The party whose claim it is: the seller, or a surety on its behalf	Seller of asset a, or Surety S1
What it warrants	Named attested fields at a named level, not the asset in general	NOI at A3 within 5% for 12 months
Size	The buyer's value loss at a stated miss, capped; never the asset value	Loss at a 10% NOI miss, about 10% of price
Form	Fiat or stablecoin escrow, yield-bearing, in a regulated account	Escrow at a bank on the rails
Trigger	Objective only: attested value against a later mechanical truth	Bank receipts over 12 months versus attested NOI
Forfeit	Pro rata to the shortfall beyond tolerance, never all or nothing	8% miss: 3% beyond tolerance, paid to buyer
Dispute	Automatic inside the band; expert determination outside it, time-boxed	30-day determination, cost to the loser
Release	On reconciliation at term, or earlier on buyer waiver; yield to the poster	Month 12, or on first audited accounts

Everything in the rule column exists today as a warranty, a holdback, an escrow and a dispute clause negotiated deal by deal. What changes is that the fields are machine-readable, the trigger is measured rather than argued, and the terms are standard enough to be priced by a third party. That last property is what makes a market of it.

Sizes and terms are illustrative; the shape is the point.

LAYER 3 · WHAT CAN BE SLASHED

Slashing needs a later truth a contract can read

OBJECTIVE · BONDABLE

- Attested rent against bank receipts
- Vacancy against lease register and receipts
- In-place debt against the loan register
- Title and encumbrances against the registry
- Closing funds against the closing set
- Service charge recovery against the ledger
- Arrears against the collections record

ADJUDICATED · BONDABLE WITH A REFEREE

- Capex backlog, settled by independent inspection
- Building condition and defects
- Environmental findings
- Lease term interpretation, side agreements

Works when the referee is named in advance, time-boxed, and paid by the losing side.

SUBJECTIVE · NEVER BOND

- Valuation marks
- Location and building class
- Tenant credit opinion
- Market outlook, ERV growth
- ESG ratings

These are the professional acts. Their assurance is a signature and a licence, not capital.

The lesson from staking. Slashing for downtime and double-signing works because the chain can see both. Slashing by governance vote over disputed conduct has failed everywhere it was tried. Keep the trigger list to the left column and the referee column; the right column is where money-based assurance breaks.

DOES IT WORK · THE ANALOGUES

Every piece of the bond exists somewhere already

Mechanism	Where it works	Why it works	Where it fails
Escrow holdbacks	M&A and real estate closings	Capital already at the table	Slow release; every claim is a negotiation
W&I insurance	Share deals across Europe	Transfers a priced residual	Long tails; excludes known issues
Retention and surety	Construction contracts	Third parties price the bond	Trigger disputes on quality
Notary escrow accounts	German conveyancing	Independent holder, mechanical release	Manual instructions, email risk
Validator staking	Proof-of-stake networks	Slashing on facts the chain sees	Subjective governance slashing
Oracle dispute rounds	Data feeds, prediction markets	Bonded assertions, escalating challenge	Slow and gameable on soft facts

The pattern. Every mechanism works where the trigger is observable and fails where it is argued. The platform's contribution is making more real estate facts observable, not inventing a new form of collateral.

The nearest relative is the surety bond. A third party posts for a fee, prices by the principal's record, and recovers from the principal on a claim. That is a market, and it is the bridge to insurance.

DOES IT WORK · THE CAPITAL QUESTION

Someone has to put up capital. Who, how much, and why

WORKED EXAMPLE

illustrative

Asset price · attested NOI · cap rate €13.0M · €800k · 6.15%

Warranty: NOI within 5% over 12 months tolerance €40k

Bond sized to value loss at a 10% miss €1.3M
(€80k ÷ 6.15%)

Bond as share of price 10%

Seller's cost if self-bonded: escrow earns liquidity only
yield

Seller's cost if a surety posts: annual fee on [x]% × €1.3M
€1.3M

What the seller buys: A4 on NOI rank, firm debt, speed

Realised NOI €760k: 5% miss, inside tolerance, bond released in full.

Realised €720k: 10% miss, 5% beyond tolerance, half the bond to the buyer.

NET NEW CAPITAL IS USUALLY SMALL

Most institutional sales already carry a holdback, a retention or a W&I premium. The bond replaces those, sized the same way. Where a deal carried none, the seller is being asked for capital it would otherwise have conceded in price.

THE TEST IT HAS TO PASS

A seller posts when the certainty premium exceeds the bond's cost: a higher rank in every mandate, firm term sheets earlier, a shorter hold, fewer retrades. If the market does not pay for certainty, nobody bonds and the mechanism idles. It has to be measured, not assumed.

Surety fee left as a placeholder; it is what the surety market will price.

THE BOND MARKET

Three ways to post, and the one that becomes a market

1 · SELF-BOND

Seller posts its own escrow

Simplest, no third party, yield stays with the seller. Costs liquidity. The default for large sellers with cash and a clean record.

2 · SURETY

A third party posts for a fee

Sureties rest standing terms on the platform like credit boxes: fields covered, levels required, fee by seller record. The seller takes the cheapest quote. Fees fall with assurance level and history.

3 · POLICY

Insurance instead of capital

The buyer holds a policy that pays on the same trigger. No escrow at all; the insurer carries the risk against a premium. Surety and policy converge into the same underwriting engine.

The market is in option 2. Several sureties quoting the same bond on the same attested record is a book: comparable terms, one axis of price, resting intents. It is the debt-book pattern from the mandate mechanics, applied to assurance.

Who the sureties are. Insurers and specialty finance today; the platform's own underwriting entity once it has loss data. That is where the native insurer question actually enters, and the next slides take it up.

INSURANCE · THE NATIVE INSURER QUESTION

Three ways to be an insurer, only one of them early

Path	What we are	Our capital	Licence	What we learn
A · Data partner	Evidence pack and levels supplied to W&I and title insurers	None	None	Which fields move their premium
B · MGA	Delegated underwriting authority; a fronting carrier and reinsurers hold the risk	Minimal	Intermediary	Loss ratios on our own pricing
C · Own carrier	Captive or licensed insurer on our balance sheet	Solvency capital	Full, per domicile	Float; full margin; full risk

THE REALISTIC ROUTE IS A THEN B

An MGA is the underwriting engine without the balance sheet: we price, select and administer; a carrier's capital stands behind the policy; reinsurance caps our exposure. It is how most specialty lines are born, and it earns the loss data that path C needs.

WHY NOT C FIRST

A carrier is regulated in every jurisdiction that matters, holds capital against every policy, and is priced by regulators and rating agencies on loss history it does not yet have. Whitepaper-era "native insurer on chain" designs skipped this; the capital rules did not.

INSURANCE · THE UNDERWRITING ENGINE

Every deal underwritten is a policy priced later

01

Underwrite

Prop.com underwrites a deal: attested fields, levels, model outputs.

02

Snapshot

State at close is frozen with lineage: what was claimed, at what level.

03

Observe

Realised NOI, cash, arrears, claims and disputes accrue on the same record.

04

Curve

Loss frequency and severity by asset type, level, seller record and region.

05

Price

Premiums and surety fees set from the curve; sharper with every closed deal.

The edge is observation, not volume. A traditional insurer sees a deal once, at binding. The platform sees the claimed state and the realised state on the same asset for years. That pairing is the loss curve; no incumbent has it.

The lag is real. W&I claims arrive over years; a curve needs cycles, not quarters. Start with short-tail products where the outcome lands in twelve months, and let the long-tail curves build underneath.

Short tails first, because they teach fastest

Product	Trigger	Tail	Data it needs	Buyer
Data warranty cover	Realised NOI misses attested NOI beyond tolerance	12 months	A3 history, cash rails	Buyer, lender
Surety bond	Seller's bond is forfeited	12 months	Seller record, levels	Seller
Settlement cover	Funds diverted or settlement fails to complete	Weeks	Closing sets, rails	Both sides
Rent default cover	Tenant arrears beyond a threshold	1 to 3 years	Ledger, tenant credit	Landlord, lender
W&I on share deals	Breach of warranty discovered after close	Up to 7 years	DD findings, full record	Buyer
Title cover	Defect in title or undisclosed encumbrance	Open-ended	Registry mirror	Lender, buyer

The first three are parametric. The trigger is a number the platform already measures, so no loss adjuster, no dispute over quantum, and a loss curve within a year. They are also the ones no incumbent can write without our data.

The last three are the established market. Enter them as a data partner and MGA, not a carrier: the tails are long, the capital is heavy, and incumbents already have the curves. Our advantage there is selection, not pricing.

Tail lengths indicative; product design subject to regulatory review per domicile.

Float is a consequence of underwriting, not a strategy

PREMIUMS IN

Collected at binding, before any claim exists.



RESERVES HELD

Invested until claims fall due. This is the float; its value is duration times return.



CLAIMS OUT

Paid on trigger. Profit is premium minus claims minus cost, plus what the float earned.

WHAT IS TRUE

Float is real and long-tail lines generate a lot of it. An underwriter that loses less than the market on the same risk earns twice: on the combined ratio and on the float. Data that predicts loss is exactly the thing that makes float safe to hold.

WHAT IS ALSO TRUE

Float sits behind solvency capital the regulator sets, invested under rules the regulator sets. Short-tail parametric lines, the ones we can price first, generate little of it. The float argument is a path C argument, years out, and it only pays if the underwriting is right.

So the order of value is: the loss curve first, because it is the moat; the MGA margin second, because it is earned on someone else's capital; the float last, because it arrives only with a balance sheet and only rewards discipline.

BEING REALISTIC

What holds, and what the white papers got wrong

HOLDS

- Bonds on facts with a cash truth, sized to value at risk
- Sureties as a resting-order market on the platform
- Assurance levels as parameters mandates and boxes require
- Cash reconciliation as the arbiter of source data
- Parametric short-tail cover as the first products
- The MGA route to underwriting on a carrier's capital
- Claimed-versus-realised state as a loss curve nobody else has

DOES NOT HOLD

- Tokens as bonds: institutions will not stake them
- Slashing opinions: valuations, class, credit views
- Consensus deciding truth: it orders, it does not verify
- A chain-native carrier without licence and solvency capital
- A pricing edge before loss data exists
- Bonds larger than the holdbacks sellers already accept
- Any layer stopping a conspiracy across every source

The two open risks. Adverse selection: if bonding costs more than the certainty premium, only sellers with nothing to hide leave and the rest bond, which is fine, or only the honest leave and the rest bond cheaply, which is not; the surety fee has to reflect record, not just level. And adjudication: the referee for the middle column has to exist, be named, and be paid by the loser before the first dispute, not after.

THE PLAN ON ONE PAGE

Measure first. Bond what is measured. Insure the rest.

NOW

- Assurance levels into the engine
- Cash reconciliation via payment ops
- Standard bond contract, self-bond only
- Evidence pack to one W&I insurer

Decide: domicile and counsel for the MGA route.

NEXT

- Surety book: standing terms on the platform
- MGA with a fronting carrier
- Data warranty and settlement cover
- Named referee for adjudicated triggers

Decide: whether Prop.com or Propchain holds the underwriting entity.

LATER

- Own carrier if loss ratios earn it
- Long-tail lines: W&I, title
- Float as a balance-sheet business
- Levels as an industry standard

Decide: who the first referee is for adjudicated triggers.

Discussion

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